

15.0 Summative Assessment: Routing, Network Segmentation, and Network Tracing

To complete this summative assessment, choose one of the labs in the competency. Include detailed documentation of your configurations, troubleshooting steps, and results.

Remember:

- Keep detailed notes during each lab to make documenting easier. ✓
- Include screenshots or configuration files to support your documentation. ✓
- Reflect on what you learned in each lab and any challenges you faced.
 - I chose this lab because it guided me with detailed steps on securing network devices. I appreciated that I had to search for and verify the correct steps and information for security best practices, which helped me get very familiar with using commands to manage routers and switches and configure secure devices.
 - YouTube Video Walkthrough: [Lab 21 - Secure Network Devices \(YouTube\)](#)
 - Lab Walkthrough: [Lab 21| Cisco Networking Labs - Secure Network Devices](#)

Packet Tracer - Lab 21 Secure Network Devices

Addressing Table

Device	Interface	Address	Mask	Gateway
RTR-A	G0/0/0	192.168.1.1	255.255.255.0	N/A
	G0/0/1	192.168.2.1	255.255.255.0	N/A
SW-1	SVI	192.168.1.254	255.255.255.0	192.168.1.1
PC	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Laptop	NIC	192.168.1.10	255.255.255.0	192.168.1.1
Remote PC	NIC	192.168.2.10	255.255.255.0	192.168.2.1

Requirements

Note: To keep this activity brief and easy to manage, some security configuration settings have not been made. In other cases, security best practices have not been followed.

In this activity you will configure a router and a switch based on a list of requirements.

Instructions

Step 1: Document the Network

Complete the addressing table with the missing information.

Step 2: Router configuration requirements:

1. Prevent IOS from attempting to resolve mistyped commands to domain names.
 - Router>enable
 - Router#show running-config
 - Router#configure terminal
 - i. Enter configuration commands, one per line. End with CNTL/Z.
 - Router(config)#no ip domain-lookup
2. Hostnames that match the values in the addressing table.
 - Router(config)#hostname RTR-A
3. Require that newly created passwords be at least 10 characters in length.
 - RTR-A(config)#security passwords min-length 10
4. A strong ten-character password for the console line. Use @Cons1234!
 - RTR-A(config)# line console 0
 - RTR-A(config-line)# password @Cons1234!
 - RTR-A(config-line)# login
5. Ensure that console and VTY sessions close after 7 minutes exactly.
 - RTR-A(config-line)#exec-timeout 7 0
 - RTR-A(config-line)#exit
 - —
 - RTR-A(config)#line vty 0 15
 - RTR-A(config-line)#exec-timeout 7 0
 - RTR-A(config-line)#exit
6. A strong, encrypted ten-character password for the privileged EXEC mode. For this activity, it is permissible to use the same password as the console line.
 - RTR-A(config)# enable secret @Cons1234!
7. A MOTD banner that warns about unauthorized access to the devices.
 - RTR-A(config)#banner motd 'Username / Password Required!'
8. Password encryption for all passwords.
 - RTR-A(config)# service password-encryption
9. A user name of NETadmin with encrypted password LogAdmin!9.
 - RTR-A(config)#username NETadmin secret LogAdmin!9
10. Enable SSH.
 - Use security.com as the domain name.
 - i. RTR-A(config)# ip domain-name security.com
 - ii. RTR-A(config)# crypto key generate rsa
 - Use a modulus of 1024.
 - i. When prompted: How many bits in the modulus [512]: 1024
 - ii. RTR-A(config)# ip ssh version 2

11. The VTY lines should use SSH for incoming connections.
 - o **RTR-A(config)# line vty 0 15**
 - o **RTR-A(config-line)# transport input ssh**
12. The VTY lines should use the username and password that were configured to authenticate logins.
 - o **RTR-A(config-line)# login local**
 - o **RTR-A(config-line)# exit**
13. Impede brute force login attempts by using a command that blocks login attempts for 45 seconds if someone fails three attempts within 100 seconds.
 - o **RTR-A(config)# login block-for 45 attempts 3 within 100**
 - o **RTR-A(config)# exit**
 - o **RTR-A#show running-config**
 - o **RTR-A#copy running-config startup-config**
 - i. Destination filename [startup-config]?
 - ii. Building configuration...
 - iii. [OK]
 - o **RTR-A#**

Step 3: Switch configuration requirements:

1. Assign hostname, assign banner and prevent IOS from attempting to resolve mistyped commands to domain names.
 - a. **Switch>enable**
 - b. **Switch#show running-config**
 - c. **Switch#configure terminal**
 - i. **Enter configuration commands, one per line. End with CNTL/Z.**
 - d. **Switch(config)#hostname SW-1**
 - e. **SW-1(config)#banner motd 'Username / Password Required!'**
 - f. **SW-1(config)# no ip domain-lookup**
2. All unused switch ports are administratively down.
 - a. **SW-1# show interfaces status**
 - b. **SW-1(config)# interface range fa0/1-1**
 - c. **SW-1(config-if-range)# shutdown**
 - d. **SW-1(config-if-range)# exit**
 - e. **SW-1(config)# interface range fa0/3-9**
 - f. **SW-1(config-if-range)# shutdown**
 - g. **SW-1(config-if-range)# exit**
 - h. **SW-1(config)# interface range fa0/11-24**
 - i. **SW-1(config-if-range)# shutdown**
 - j. **SW-1(config-if-range)# exit**
 - k. **SW-1(config)# interface range gi0/2**
 - l. **SW-1(config-if-range)# shutdown**
 - m. **SW-1(config-if-range)# exit**
3. The SW-1 default management interface should accept connections over the network.
 - a. **SW-1(config)# interface vlan 1**

- b. SW-1(config-if)# no shutdown**
- 4. Use the information shown in the addressing table. The switch should be reachable from remote networks.
 - a. SW-1(config-if)# ip address 192.168.1.254 255.255.255.0**
 - b. SW-1(config-if)# exit**
 - c. SW-1(config)# ip default-gateway 192.168.1.1**
- 5. Use **@Cons1234!** as the password for the privileged EXEC mode.
 - a. SW-1(config)# enable secret @Cons1234!**
 - i. Not specified in the lab (encrypt all passwords, require password length, and secure console, add timeout)
 - 1. SW-1(config)# service password-encryption**
 - 2. SW-1(config)# line console 0**
 - 3. SW-1(config-line)# password @Cons1234!**
 - 4. SW-1(config-line)# login**
 - 5. SW-1(config-line)#exec-timeout 7 0**
 - 6. SW-1(config-line)# exit**
- 6. Configure SSH as was done for the router.
 - a. Use security.com as the domain name.
 - i. SW-1(config)# ip domain-name security.com**
 - ii. SW-1(config)# crypto key generate rsa**
 - b. Use a modulus of 1024.
 - i. When prompted: How many bits in the modulus [512]: **1024**
 - ii. SW-1(config)# ip ssh version 2**
- 7. Create a user name of **NETadmin** with an encrypted secret password **LogAdmin!9**
 - a. SW-1(config)# username NETadmin secret LogAdmin!9**
- 8. The VTY lines should only accept connections over SSH.
 - a. SW-1(config)# line vty 0 15**
 - b. SW-1(config-line)# transport input ssh**
- 9. The VTY lines should only allow the network administrator account to access the switch management interface.
 - a. SW-1(config-line)# login local**
 - b. SW-1(config-line)#exec-timeout 7 0**
 - c. SW-1(config-line)# exit**
 - d. SW-1(config)# exit**
 - e. SW-1#show running-config**
 - f. SW-1#copy running-config startup-config**
 - i. Destination filename [startup-config]?
 - ii. Building configuration...
 - iii. [OK]
 - g. SW-1#**
- 10. Hosts on both LANs should be able to ping the switch management interface.
 - a. ping 192.168.1.254**